Experiment 6: Fuzzy Relations and Max-Min Composition

Question

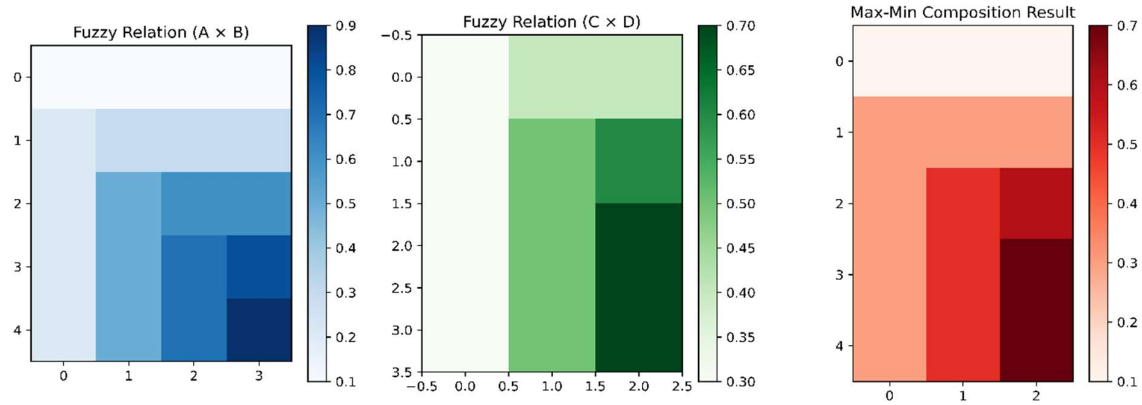
To create a fuzzy relation through the Cartesian product of two fuzzy sets and perform max-min composition on fuzzy relations.

Theory

- **Fuzzy Relation:** The Cartesian product of fuzzy sets A and B generates a relation matrix.
- **Second Fuzzy Relation:** The Cartesian product of fuzzy sets C and D forms another relation matrix.
- **Max-Min Composition:** The result of composition shows the highest minimum values between the two fuzzy relations.
- **Graphical Representation:** The heatmaps display the strength of fuzzy relationships and the effect of composition.

Implementation:

```
import numpy as np
import matplotlib.pyplot as plt

# Define fuzzy sets A and B
A = np.array([0.1, 0.3, 0.6, 0.8, 1.0])
B = np.array([0.2, 0.5, 0.7, 0.9])

# Compute Cartesian product-based fuzzy relation
fuzzy_relation = np.minimum.outer(A, B)

# Define another fuzzy relation for composition
C = np.array([0.4, 0.6, 0.8, 1.0])
D = np.array([0.3, 0.5, 0.7])

second_relation = np.minimum.outer(C, D)

# Compute max-min composition
composition_result = np.maximum.reduce(np.minimum(fuzzy_relation[:, :, None],
second_relation[None, :, :]), axis=1)

# Plot fuzzy relations
```

```
fig, axes = plt.subplots(1, 3, figsize=(15, 5))

# Plot first fuzzy relation
im1 = axes[0].imshow(fuzzy_relation, cmap='Blues', interpolation='nearest')
axes[0].set_title('Fuzzy Relation (A × B)')
fig.colorbar(im1, ax=axes[0])

# Plot second fuzzy relation
im2 = axes[1].imshow(second_relation, cmap='Greens', interpolation='nearest')
axes[1].set_title('Fuzzy Relation (C × D)')
fig.colorbar(im2, ax=axes[1])

# Plot max-min composition result
im3 = axes[2].imshow(composition_result, cmap='Reds', interpolation='nearest')
axes[2].set_title('Max-Min Composition Result')
fig.colorbar(im3, ax=axes[2])
plt.show()
```

Output